

Md. Robiul Islam Roman

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering graduate from AIUB, currently pursuing a Master of Engineering (M.Engg.) in Computer Science & Engineering at BRAC University. Proficient in MATLAB, Python, C/C++, and SystemVerilog, with hands-on experience in real-time data acquisition, biomedical signal processing, RTL/VLSI design, and embedded systems development.

SKILLS

Programming Languages:	MATLAB, C/C++, Python, SystemVerilog, PLC, SQL
EDA & Design Tools:	Cadence Virtuoso, Xcelium, Genus, NI Multisim, CST Microwave Studio, Proteus
Hardware & Embedded:	Raspberry Pi, FPGA, Embedded Systems, Computer-Aided Design (CAD), SolidWorks
Operating Systems:	Windows, Linux

EDUCATION

Master of Engineering (M.Engg.) — Computer Science & Engineering BRAC University	April 2025 – Present
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Bachelor of Science (BSc) — Electrical and Electronic Engineering American International University–Bangladesh (AIUB) CGPA: 3.74 / 4.00	Graduated February 2026
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Higher Secondary Certificate (HSC) — Science Kurmitola High School and College, Dhaka, Bangladesh	January 2021
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EXPERIENCE

Industrial Attachment Trainee Energypac Power Generation PLC, Dhaka	Nov 2025 – Dec 2025
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- Completed hands-on industrial training on diesel generators and Automatic Transfer Switch (ATS) systems.
- Gained practical knowledge of generator operation, load handling, and ATS functionality in large-scale industrial power environments.

Private Tutor Self-Employed, Dhaka	Jan 2023 – Jul 2025
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- Tutored students across secondary and higher secondary levels, fostering strong study habits, analytical thinking, and exam readiness.
- Designed personalized learning strategies that improved students' academic confidence and performance.

CERTIFICATIONS & AWARDS

Best Capstone Project Award — Faculty of Engineering, AIUB (October 2025) - Led a team developing a Continuous Cuffless Blood Pressure Monitoring System, coordinating hardware design, embedded software, sensor integration, and testing phases. - Showcased the project at Broadband Expo 2026 (ISPAB) and an international symposium hosted by the Magna Charta Observatory. Runner-Up on Poster Competition — ROBOLUTION 2025 (November 2025)
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Dean's List Honors — Faculty of Engineering, AIUB

- Awarded for three consecutive semesters: Fall 2022 (Mar 2023), Spring 2022–23 (Feb 2024), Summer 2024–25 (Oct 2025).

Academic Scholarship — Faculty of Engineering, AIUB

- Awarded academic scholarship for outstanding academic performance from **Spring 2022–23 to Spring 2024–25**.

RTL Design, Verification, Synthesis & PnR for Digital VLSI Design — AIUB Institute of Continuing Education (ICE)

- Hands-on training covering the complete digital VLSI design flow: RTL coding, functional verification, synthesis, STA, and physical design (PnR) using industry-standard EDA tools.

Chip-Based VLSI Design for Industrial Applications — Coursera & L&T EduTech (October 2025)

- Completed specialization covering Digital Design, VLSI Chip Design (Electric VLSI EDA), VHDL Programming, and FPGA Architecture.

Data Analysis with Python — Institute of Information Technology (IIT), Jahangirnagar University

- Completed professional training in Python-based data analysis under the EDGE Project, Bangladesh Computer Council, ICT Division (**Nov 2024 – Jan 2025**).

PROJECTS

Cuffless Blood Pressure Estimation using ECG & PPG Signals

- Developed a portable, wearable device for continuous non-invasive blood pressure estimation using ECG and PPG signal fusion.

Raspberry Pi-Based Person Tracking Robot

- Built a robot with real-time face/person detection and autonomous following capability using computer vision on embedded hardware.

Physical Layout Design using Cadence Virtuoso

- Implemented and verified physical IC layouts using the Cadence Virtuoso toolchain, including DRC and LVS checks.

RTL design, synthesis, and place-and-route implementation of a RISC-V3 processor.

- Designed pipelined Developed RTL design, synthesis flow, and place-and-route implementation of a RISC-V3 processor using SystemVerilog.

MATLAB-Based Music Signal Feature Extraction

- Developed a MATLAB pipeline for extracting time-domain and frequency-domain audio features for classification tasks.

PUBLICATIONS

Development of a PPG-Based Low-Cost Cuffless Continuous Blood Pressure Monitoring with Vital Sign Detection System

- IEEE BECITHCON, November 2025.

Predicting Diabetes in Bangladesh Using Machine Learning: A Data-Driven Approach

- International Journal of Modern Engineering and Management (IJMEM), Maharashtra, India, Vol. 2, Issue 4, April 2025.

Portable Health Insights: Designing a BIA-Enabled Smartwatch for Clinical and Personal Use

- RURS 1st International Student Conference and Research Fair 2024, University of Rajshahi, February 22–23, 2025.

REFERENCES

Dr. Md. Hasan Imam

Associate Professor, Department of EEE & Computer Engineering, AIUB

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